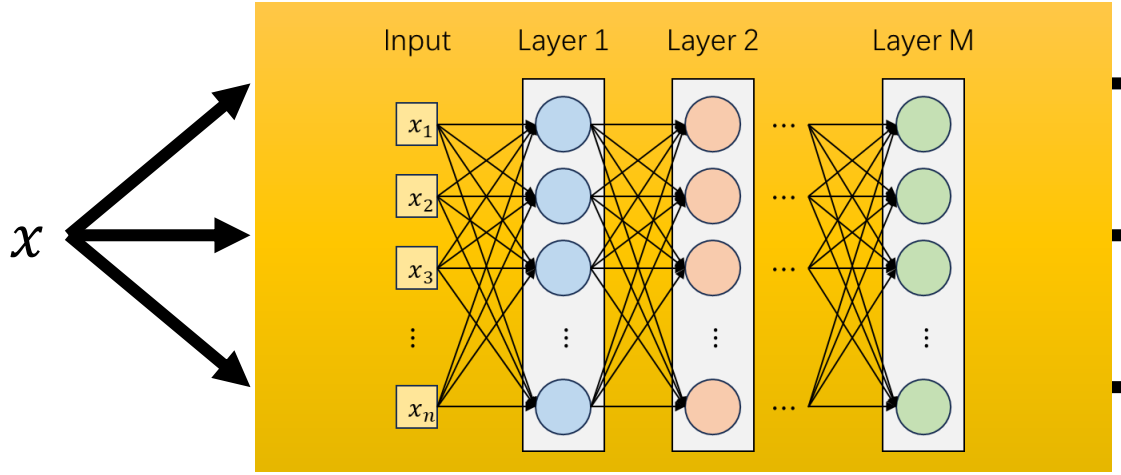
第5.2节 反向传播

训练深度神经网络的核心算法

深度学习之模型训练

Softmax回归 ➡ 深度神经网络



$\hat{y}$

$\hat{y}_1$

$\hat{y}_2$

$\hat{y}_3$

交叉熵
Cross Entropy Loss

y

y_1

y_2

y_3

$$C = -\sum_{i=1}^{3} y_i ln\hat{y}_i$$

求偏导

$$\frac{\partial C}{\partial w_i}, \frac{\partial C}{\partial b_i}$$

反向传播
Backpropagation

更新参数

$\Delta w_1,$ Δb_1

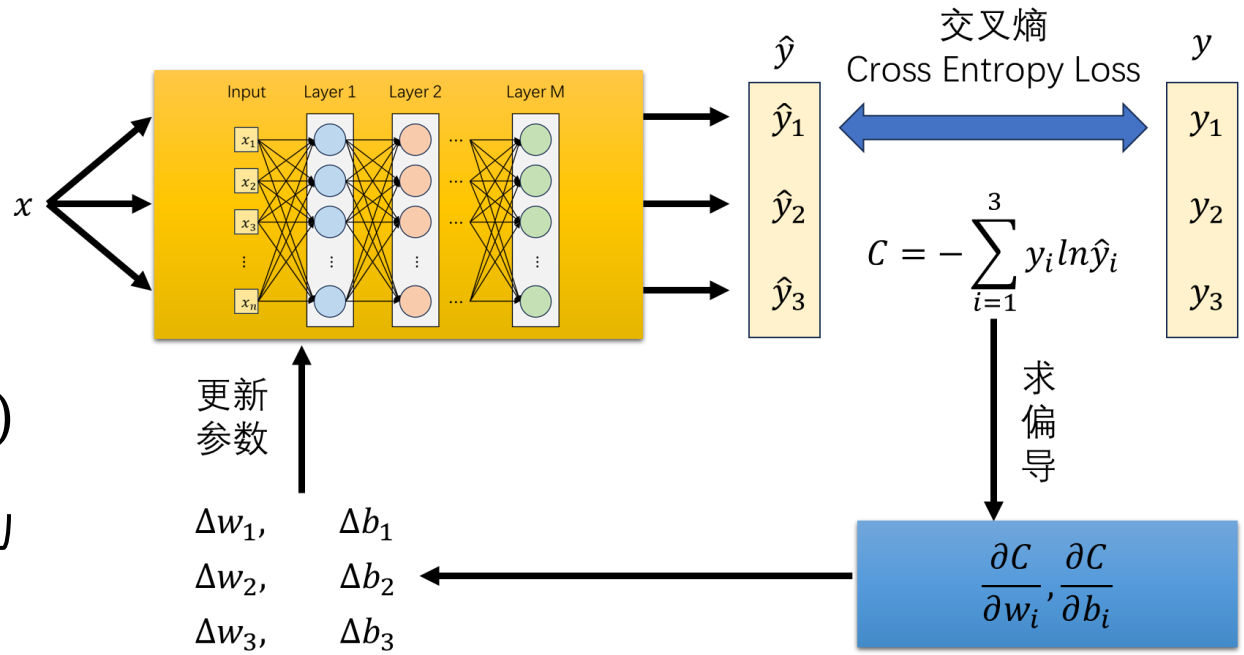
$\Delta w_2,$ Δb_2

$\Delta w_3,$ Δb_3

深度学习之模型训练——梯度下降

$$\theta = (w_1, w_2, \dots, b_1, b_2, \dots)$$

$$\Delta\theta = \underbrace{\left(\frac{\partial C}{\partial w_1}, \frac{\partial C}{\partial w_2}, \dots, \frac{\partial C}{\partial b_1}, \frac{\partial C}{\partial b_2}, \dots\right)}_{\text{维度极高}}$$



反向传播研究的问题：如何高效的计算 $\Delta\theta$ ？

链式法则

案例①: $y = g(x) \quad z = h(y)$

$$\Delta x \to \Delta y \to \Delta z \quad \Rightarrow \quad \frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx}$$

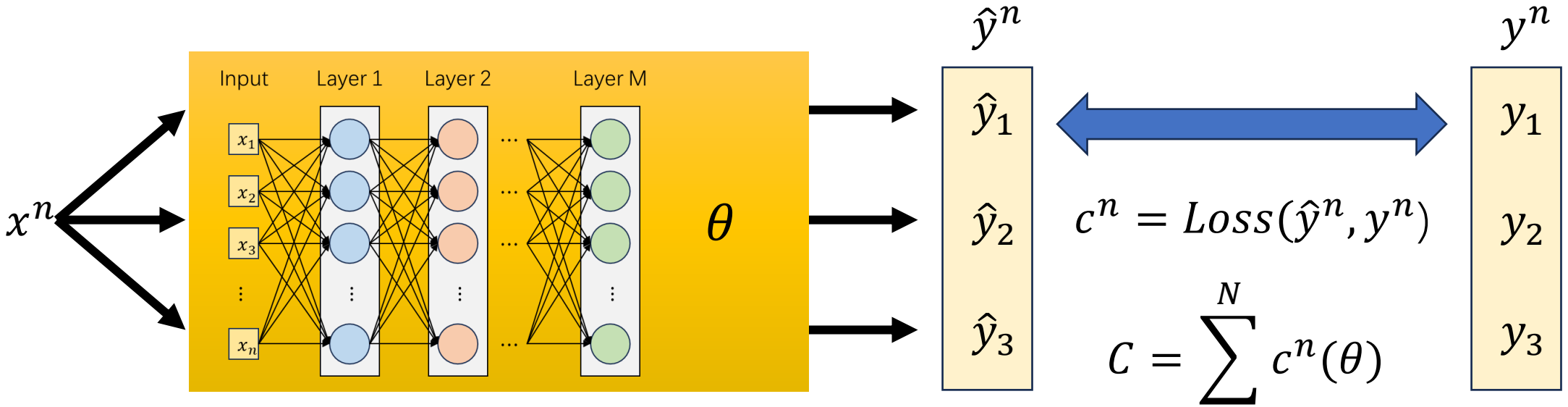
案例②: $x = g(s) \quad y = h(s) \quad z = f(x, y)$

$$\begin{tikzcd} & \Delta x \arrow[dr] & \\ \Delta s \arrow[ur] \arrow[dr] & & \Delta z \\ & \Delta y \arrow[ur] & \end{tikzcd} \quad \Rightarrow \quad \frac{dz}{ds} = \frac{\partial z}{\partial x}\frac{dx}{ds} + \frac{\partial z}{\partial y}\frac{dy}{ds}$$

案例③: $x = g(s) \quad y = h(x) \quad z = f(y)$

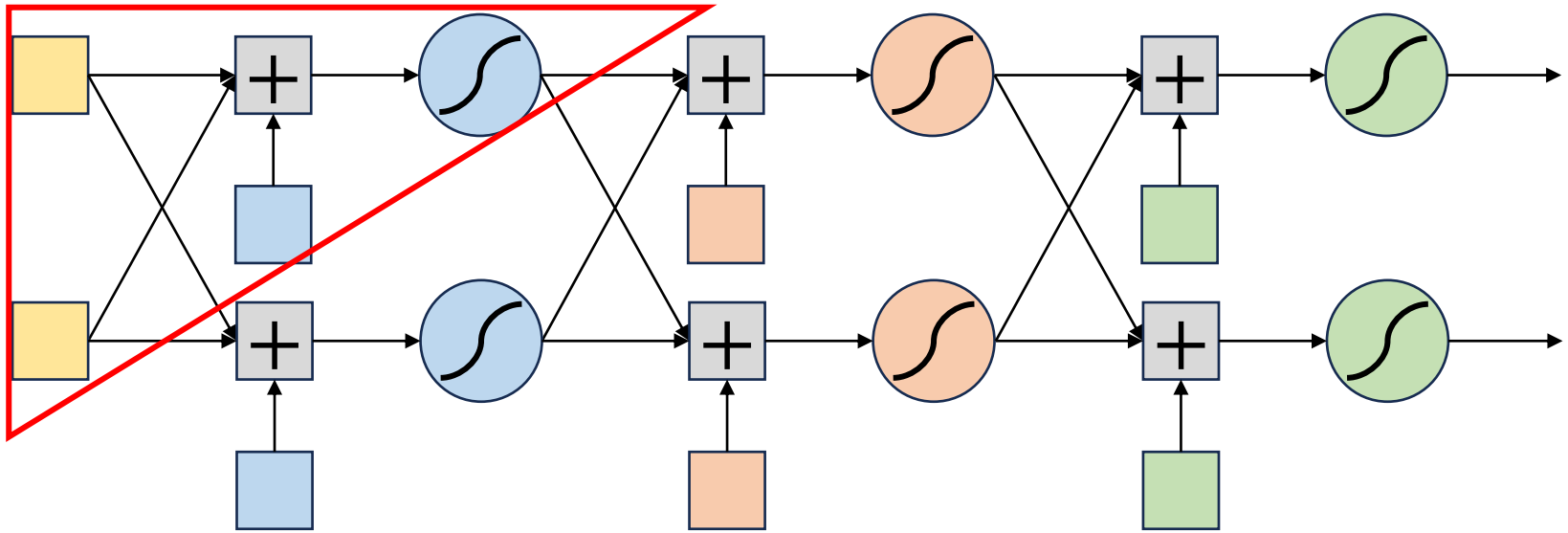
$$\Delta s \to \Delta x \to \Delta y \to \Delta z \quad \Rightarrow \quad \frac{dz}{ds} = \frac{dz}{dx}\frac{dx}{ds} = \frac{dz}{dy}\frac{dy}{dx}\frac{dx}{ds}$$

梯度计算



$$c^n = Loss(\hat{y}^n, y^n)$$

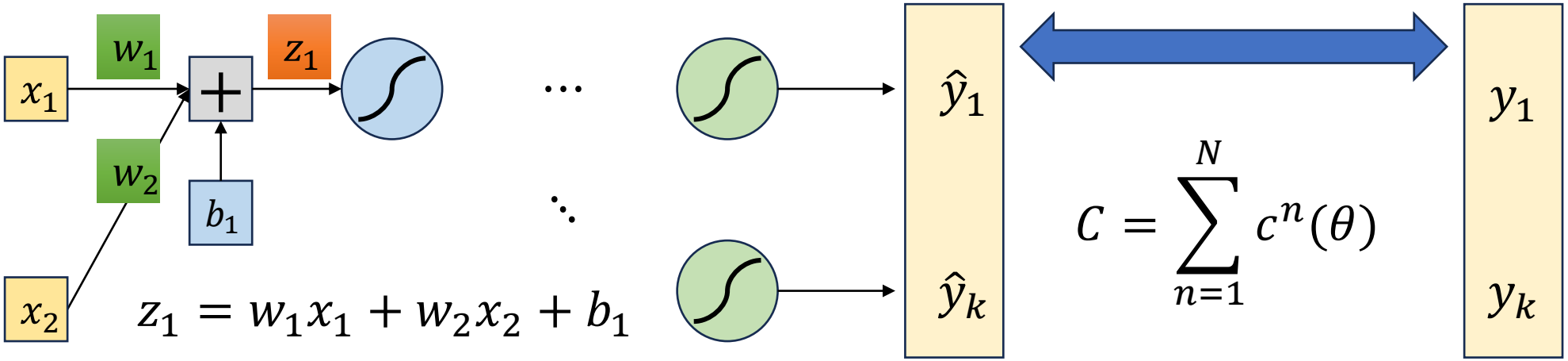
$$C = \sum_{n=1}^{N} c^n(\theta)$$



计算参数梯度

$$\frac{\partial C}{\partial w} = \sum_{n=1}^{N} \frac{\partial c^n(\theta)}{\partial w}$$

计算一个就能计算所有



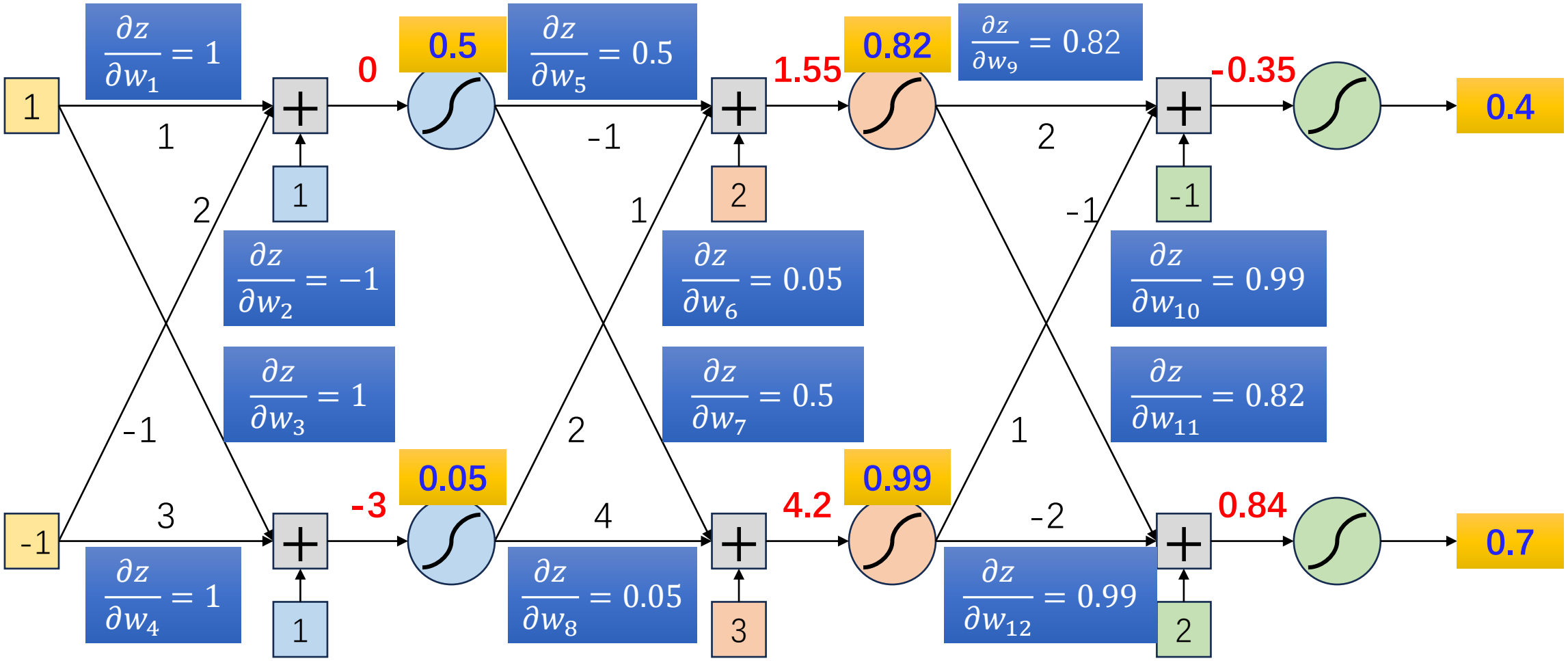
$$z_1 = w_1 x_1 + w_2 x_2 + b_1$$

$$C = \sum_{n=1}^{N} c^n(\theta)$$

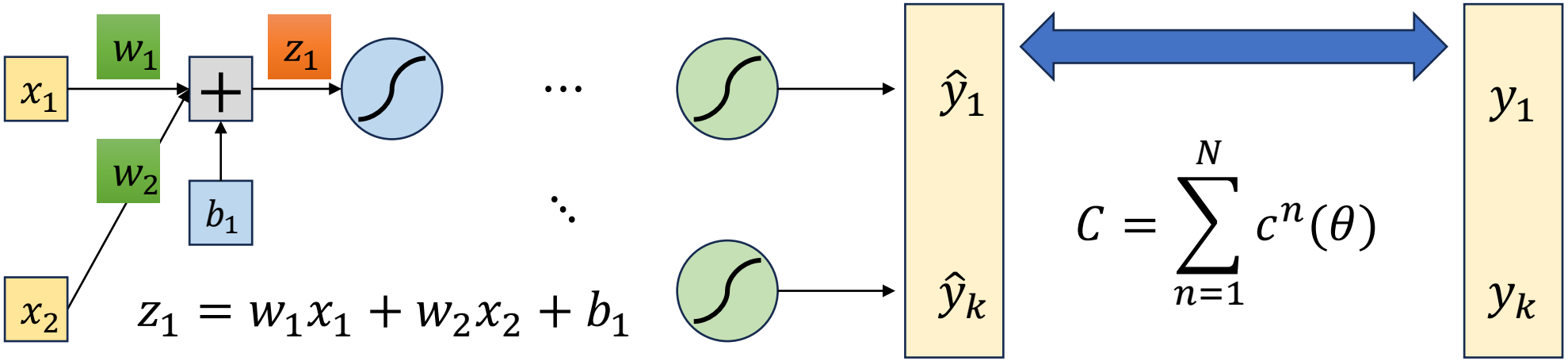
$$\frac{\partial C}{\partial w_1} = \boxed{\frac{\partial z}{\partial w_1}} \frac{\partial C}{\partial z} \qquad \frac{\partial C}{\partial w_2} = \boxed{\frac{\partial z}{\partial w_2}} \frac{\partial C}{\partial z} \qquad \frac{\partial C}{\partial b_1} = \boxed{\frac{\partial z}{\partial b_1}} \frac{\partial C}{\partial z}$$

x_1 $\qquad$ x_2 $\qquad$ 1

规律：$\frac{\partial z}{\partial w}$的值为输入的数值，$\frac{\partial z}{\partial b}$的值为1



规律：$\frac{\partial z}{\partial w}$的值可在预测时获得，即正向传播



$$z_1 = w_1 x_1 + w_2 x_2 + b_1$$

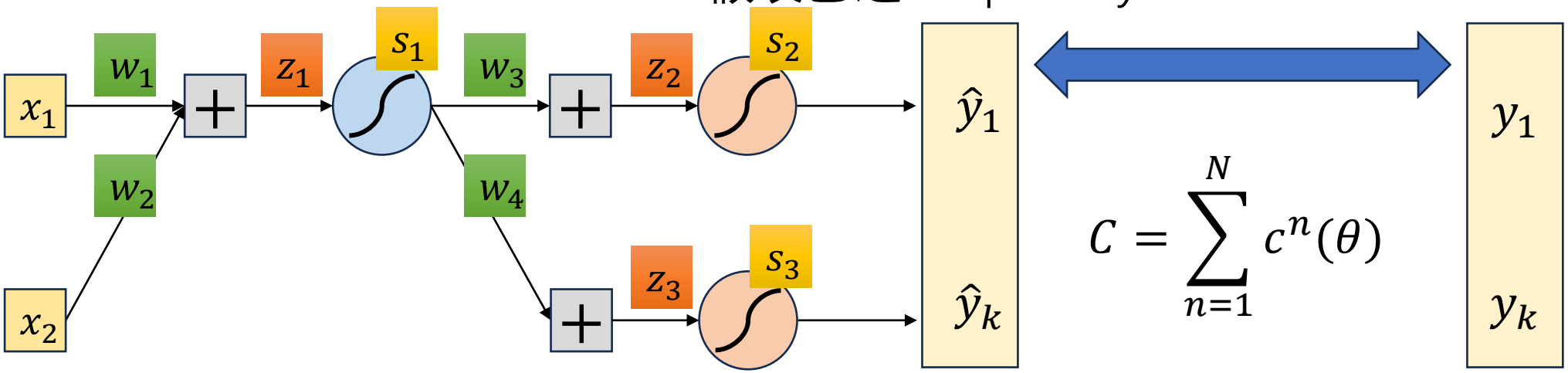
$$C = \sum_{n=1}^{N} c^n(\theta)$$

$$\frac{\partial C}{\partial w_1} = \boxed{\frac{\partial z}{\partial w_1}} \frac{\partial C}{\partial z} \qquad \frac{\partial C}{\partial w_2} = \boxed{\frac{\partial z}{\partial w_2}} \frac{\partial C}{\partial z} \qquad \frac{\partial C}{\partial b_1} = \boxed{\frac{\partial z}{\partial b_1}} \frac{\partial C}{\partial z}$$

正向传播

该如何计算 $\frac{\partial C}{\partial z}$ ？ 反向传播

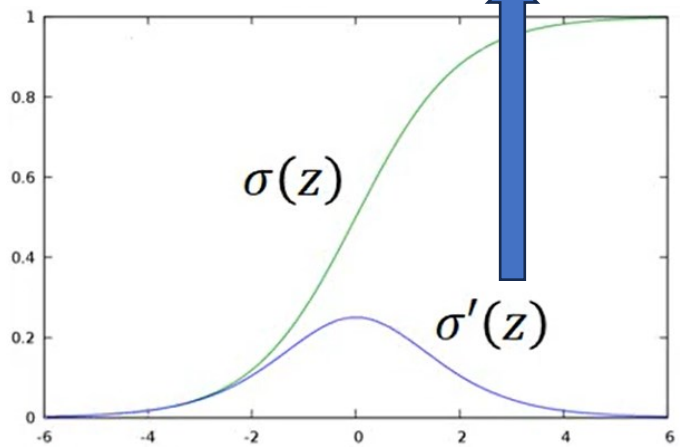
假设已达Output Layer



$$C = \sum_{n=1}^{N} c^n(\theta)$$

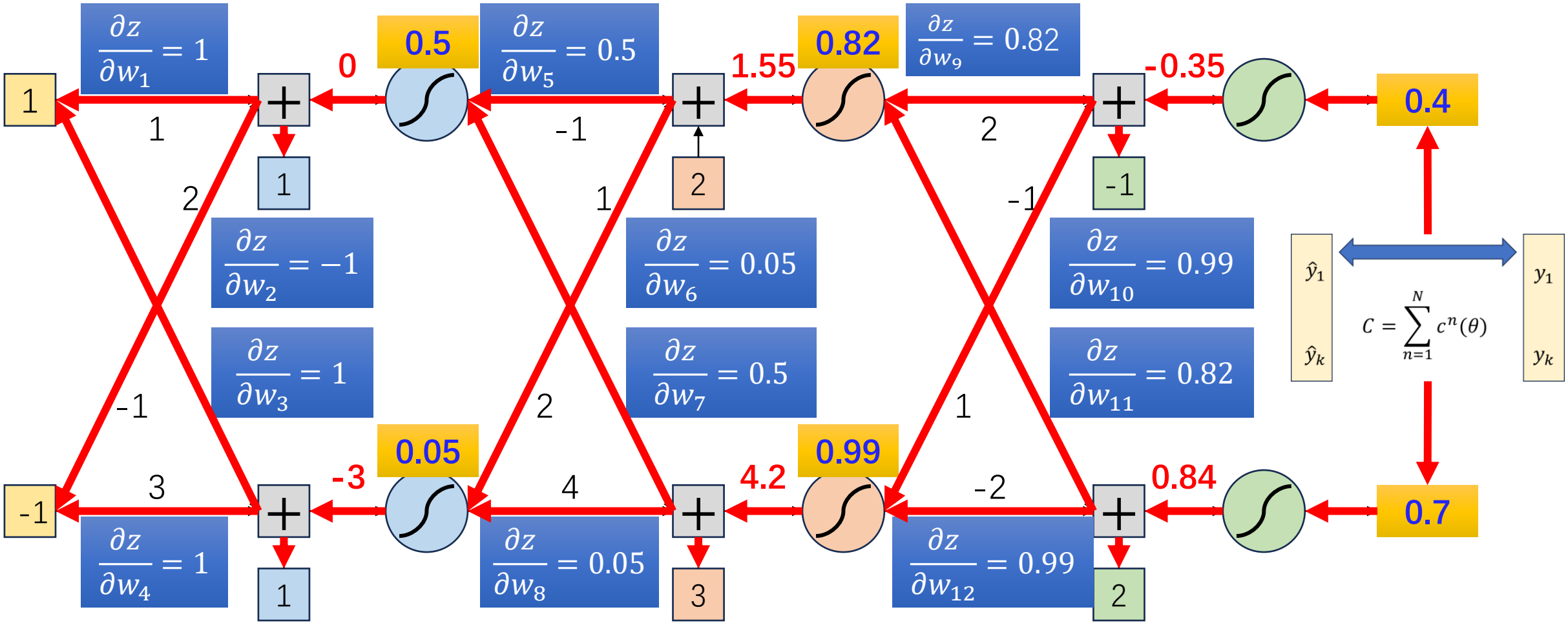
$$\boxed{\frac{\partial C}{\partial z_1}} = \frac{\partial s_1}{\partial z_1}\frac{\partial C}{\partial s_1} = \frac{\partial \sigma}{\partial z_1}\boxed{\frac{\partial C}{\partial \sigma}} \rightarrow \frac{\partial C}{\partial \sigma} = \frac{\partial z_2}{\partial \sigma}\boxed{\frac{\partial C}{\partial z_2}} + \frac{\partial z_3}{\partial \sigma}\boxed{\frac{\partial C}{\partial z_3}} \qquad \frac{\partial C}{\partial z_3} = \frac{\partial s_3}{\partial z_3}\boxed{\frac{\partial C}{\partial s_3}}$$

w_3 w_4 $\sigma'(z_3)$



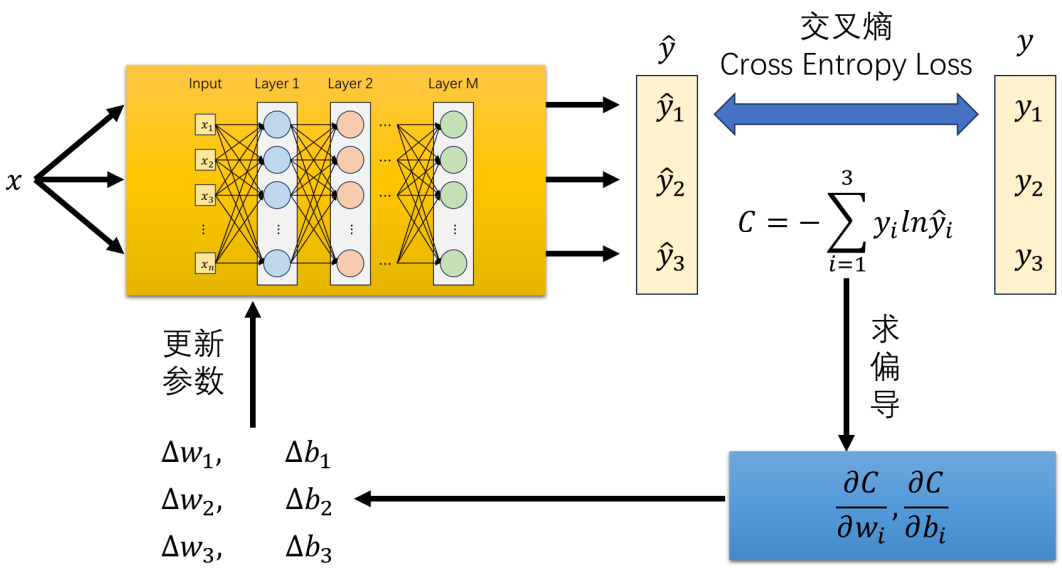
无法直接得出结果的$\frac{\partial C}{\partial z_1}$越来越多，但形式是重复的

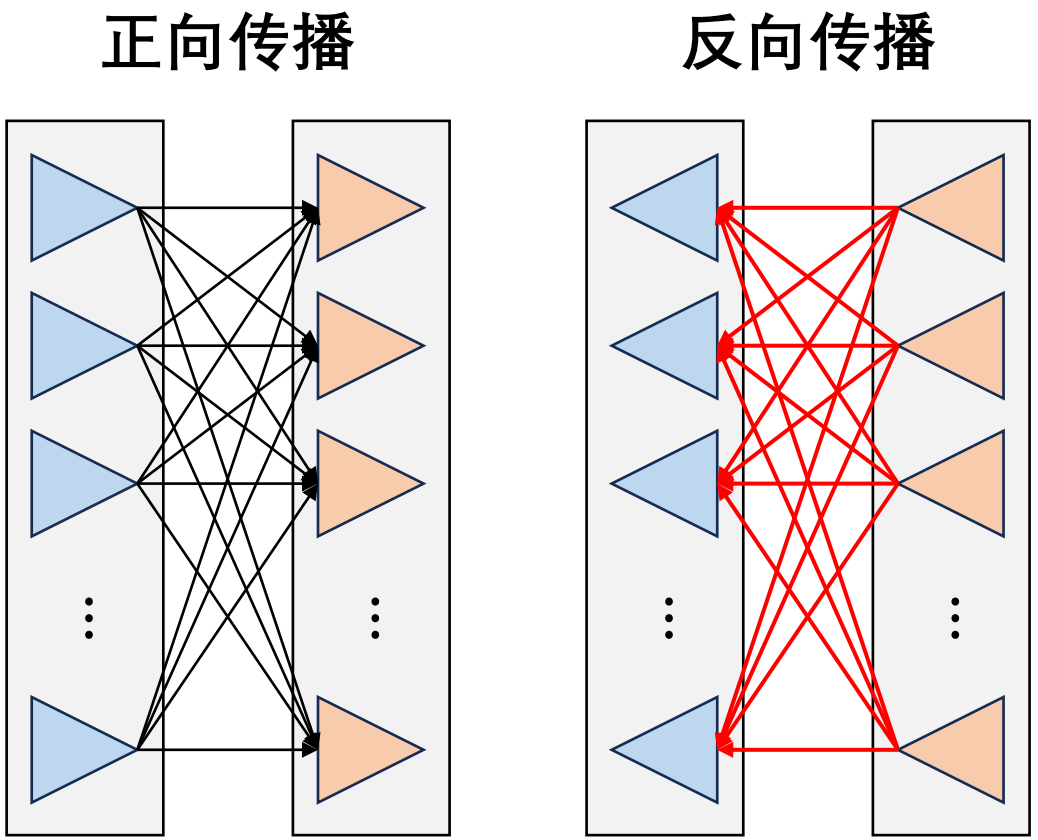
$$\frac{\partial C}{\partial s_3} = \sum_{n=1}^{N} \frac{\partial c^n(\theta)}{\partial \hat{y}_3}\Big|_{\hat{y}_3 = s_3}$$



规律：$\frac{\partial C}{\partial z}$的值需在一次预测之后，反向计算而得，即反向传播

反向传播总结



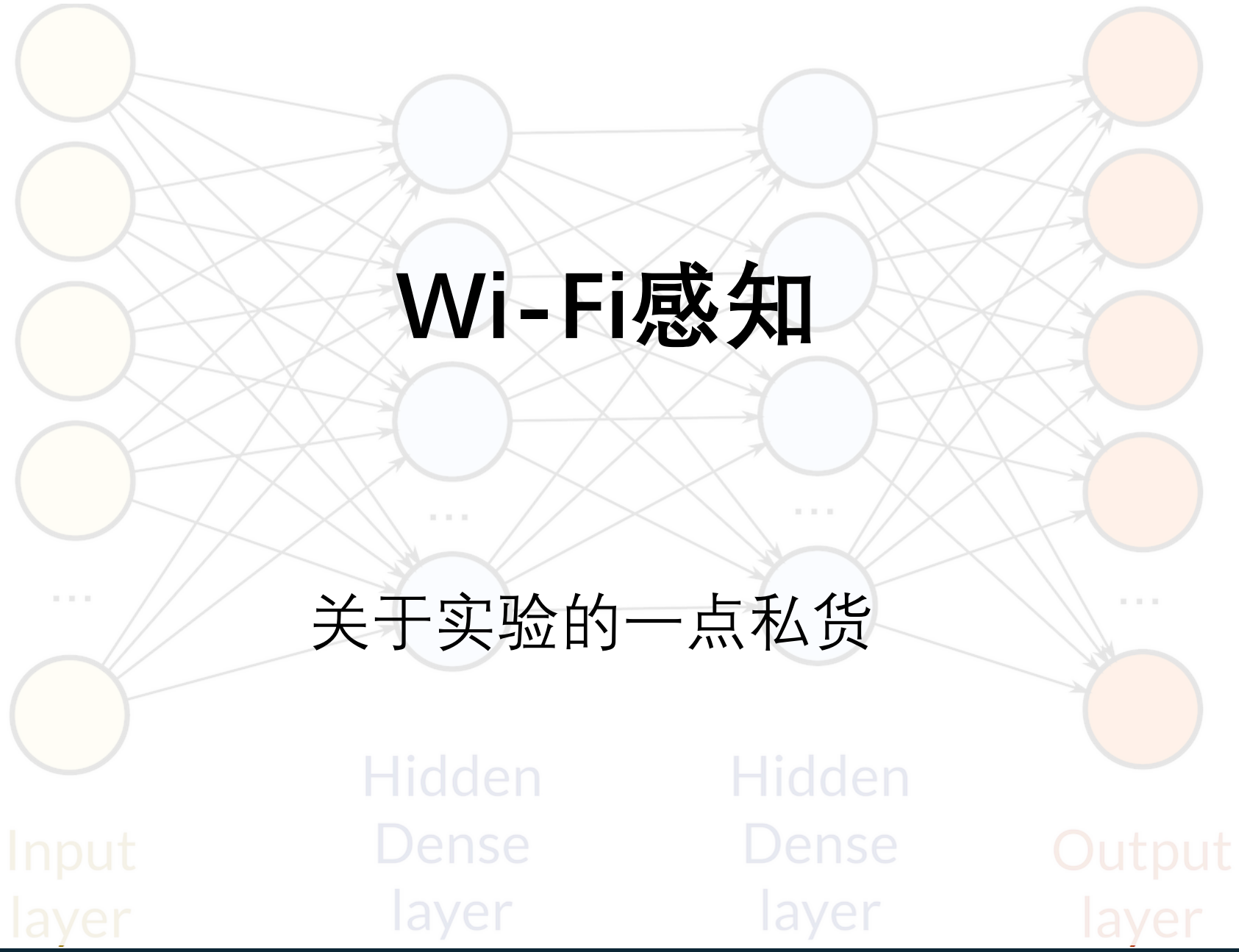


研究如何高效的计算 $\Delta\theta$ ？

$$\frac{\partial C}{\partial w} = \frac{\partial z}{\partial w} \times \frac{\partial C}{\partial z}$$

Wi-Fi感知

关于实验的一点私货



什么是无线感知

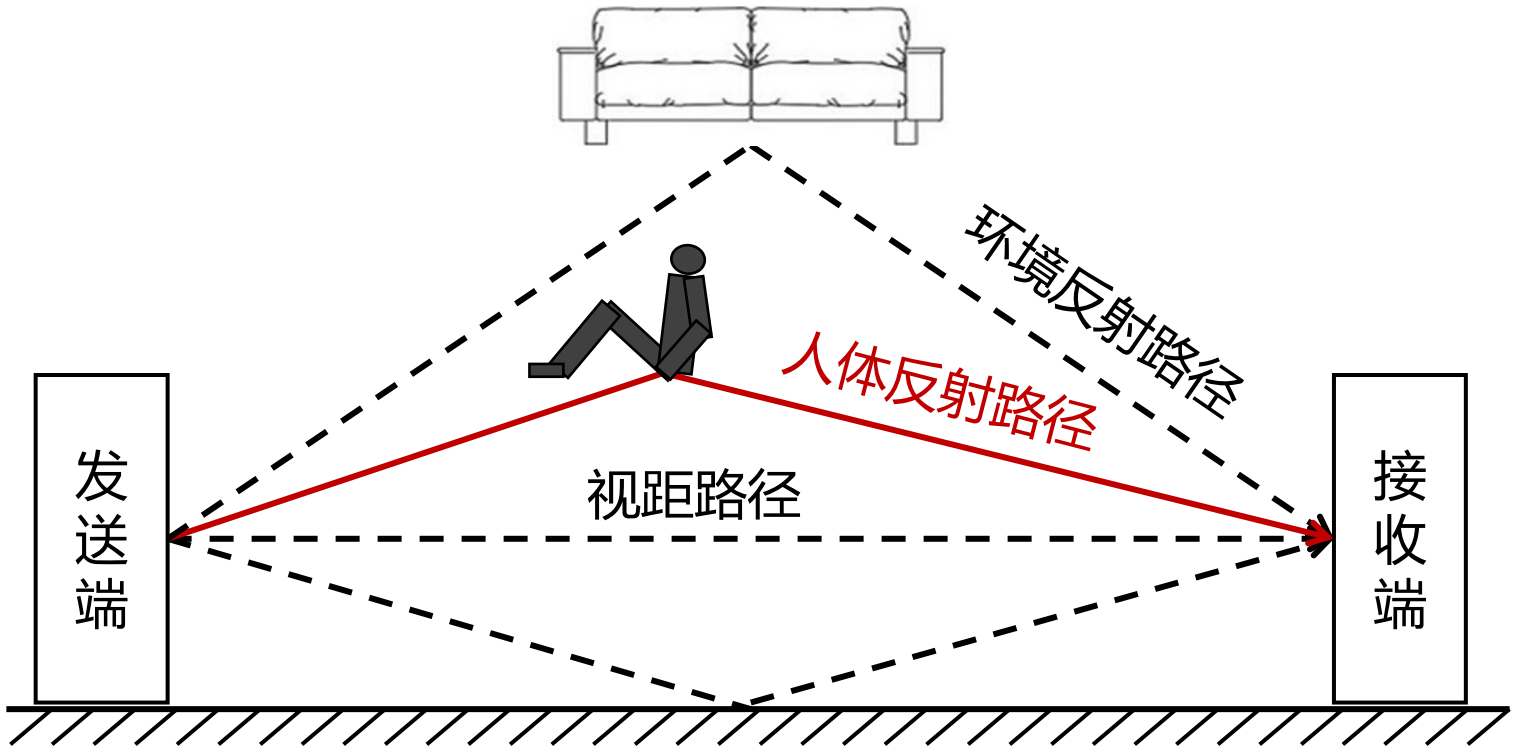
利用环境中的无线信号感知环境、物品、人类行为的技术

角色延伸——从无线通信到无线感知

前沿领域：通信感知一体化

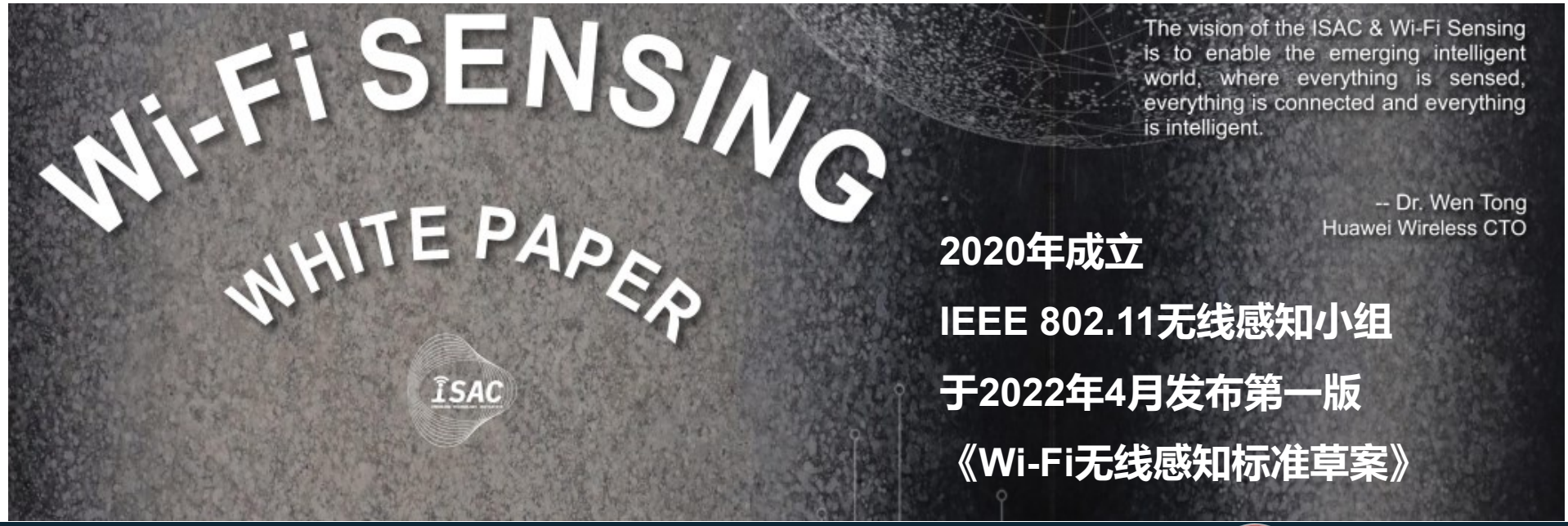
无线感知能做什么？

无线信号为什么能感知?



什么是Wi-Fi感知

- 无线感知中的一类技术
- 分析人体对Wi-Fi信号传播的干扰感知人类动作与行为
- 不受光照条件限制、无需穿戴设备、普适性强、较小的隐私担忧



2020年成立

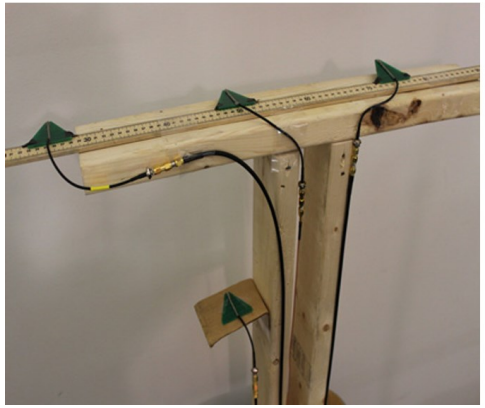
IEEE 802.11无线感知小组

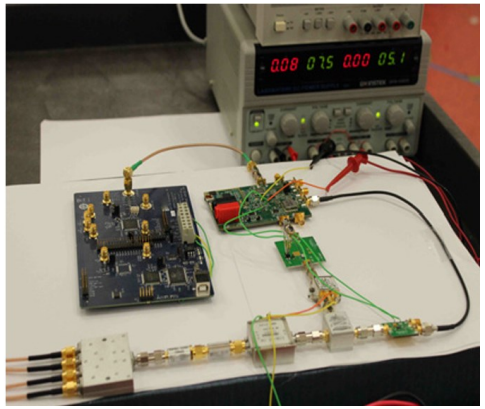
于2022年4月发布第一版

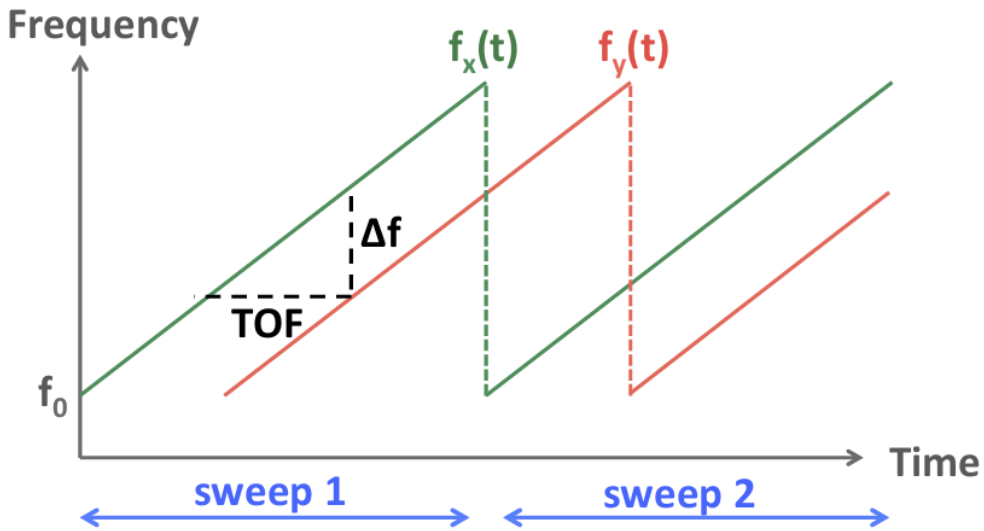
《Wi-Fi无线感知标准草案》

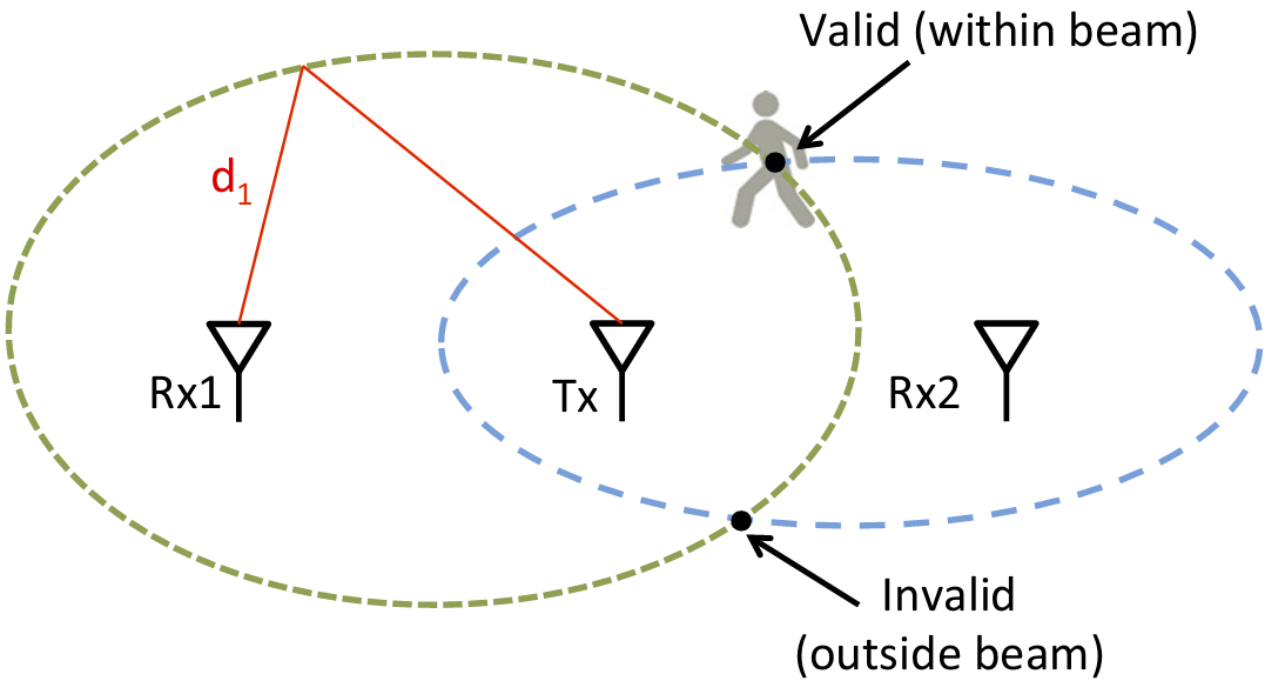
Wi-Fi感知的应用场景

室内定位



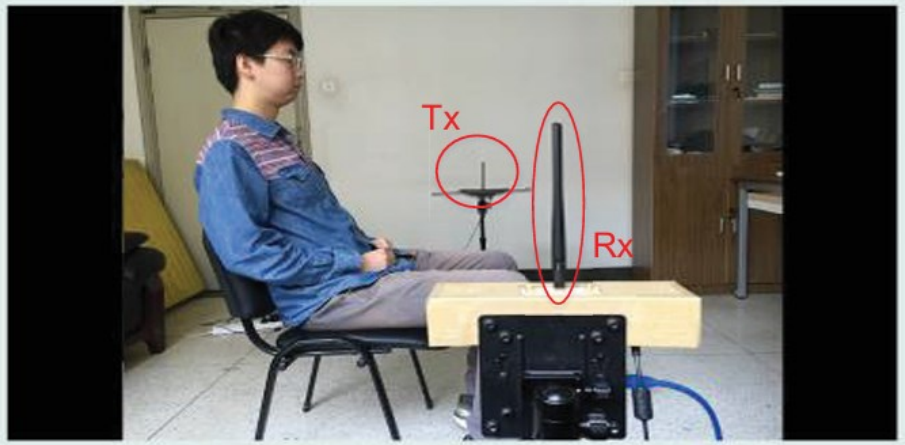




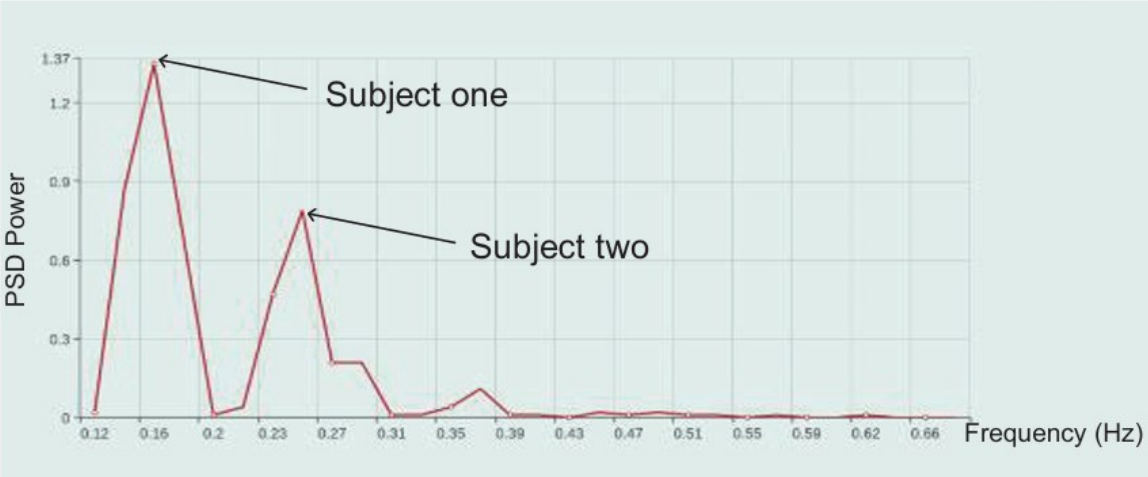


Wi-Fi感知的应用场景

呼吸监测

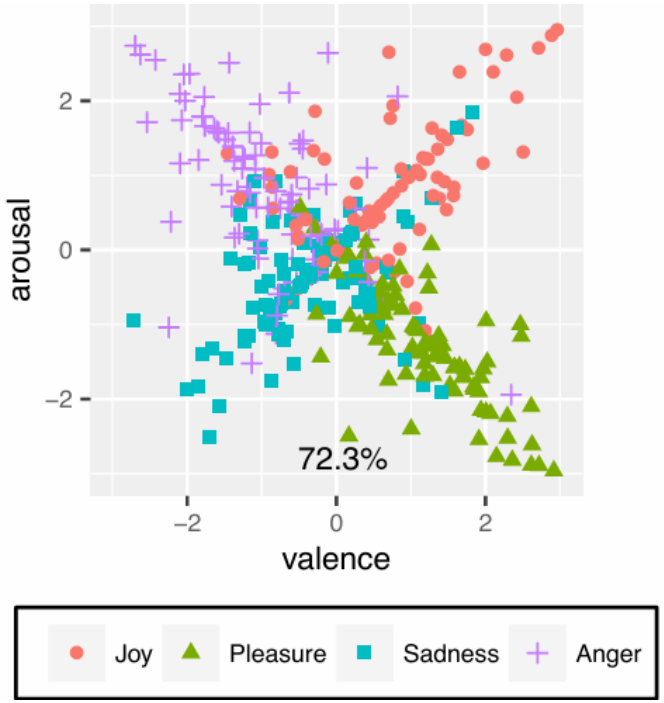


(a)

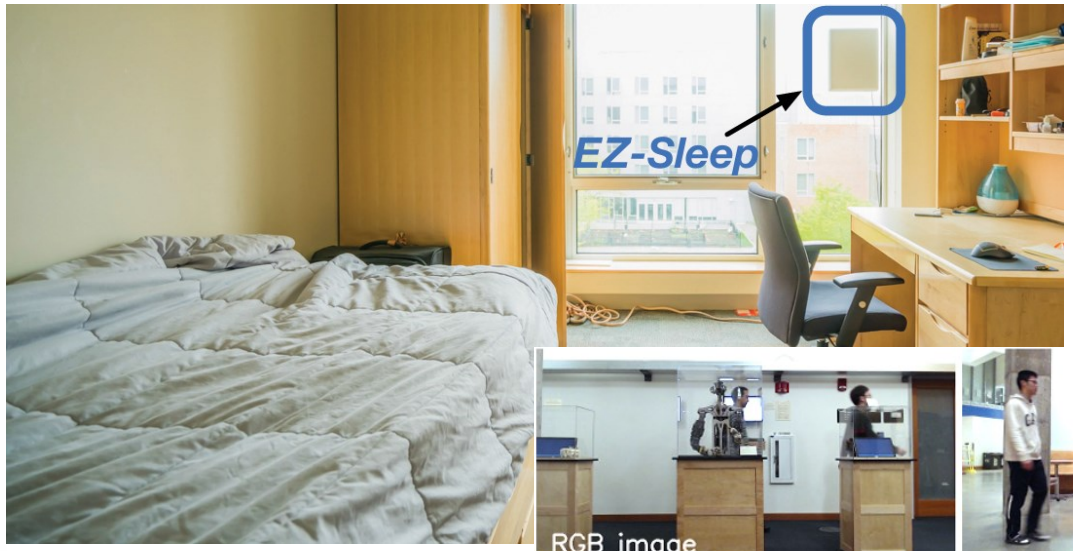


Wi-Fi感知的应用场景

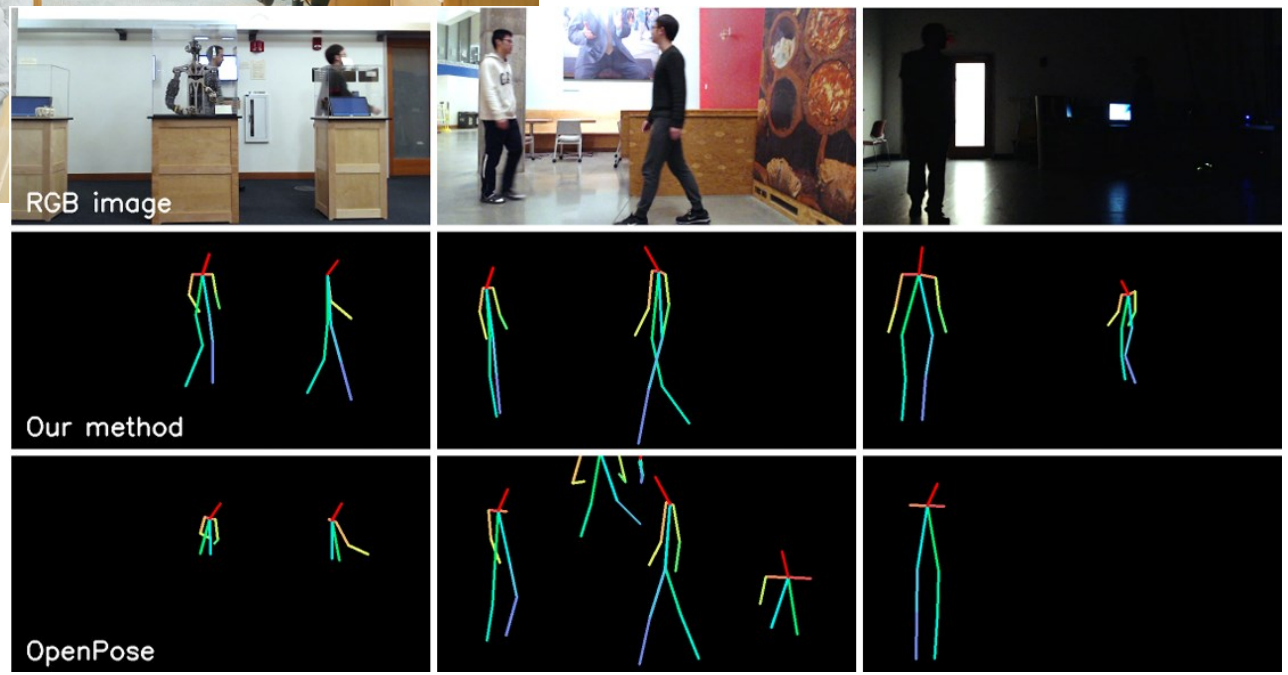
情绪识别



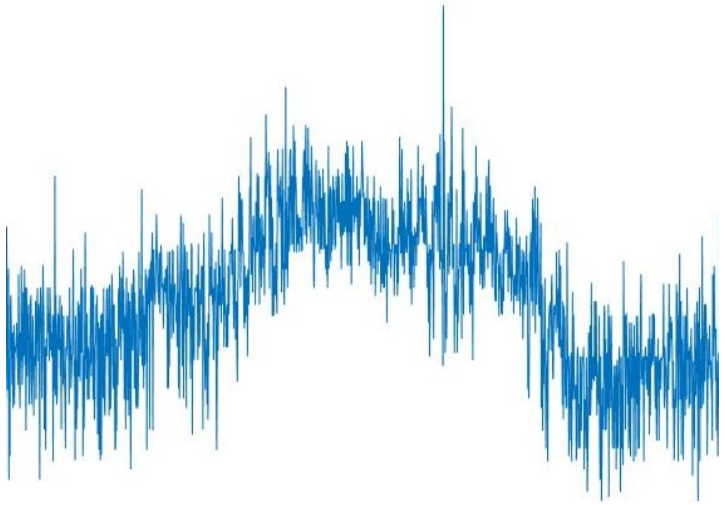
睡眠监测



动作追踪



Wi-Fi感知的应用场景

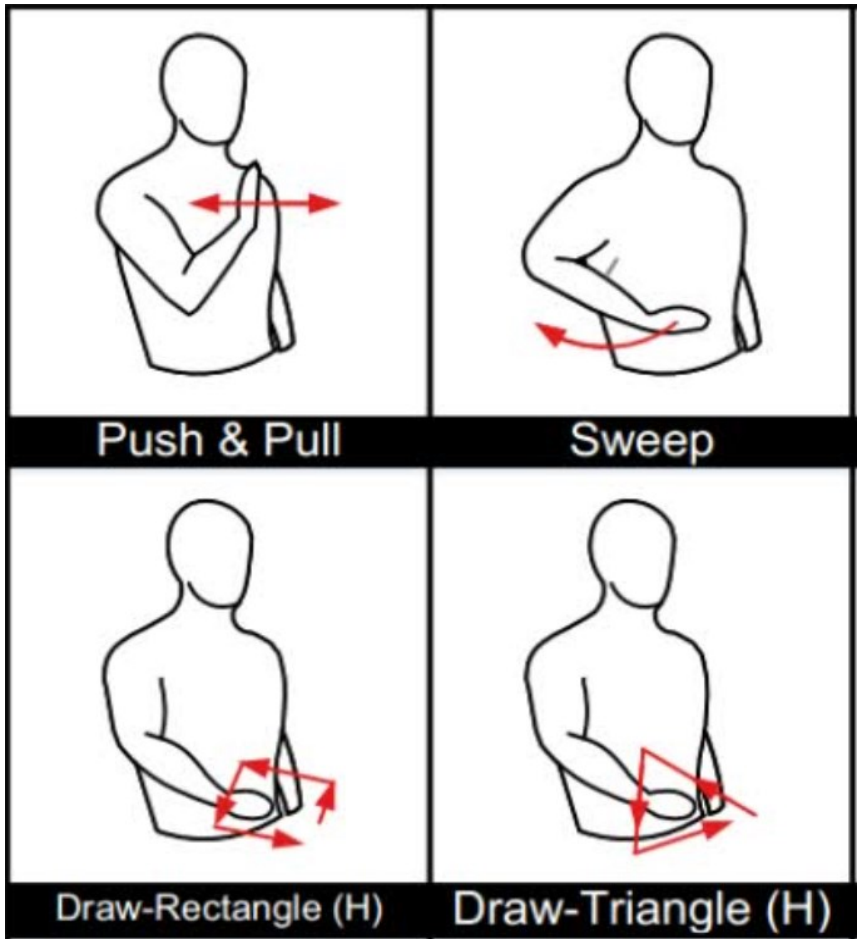


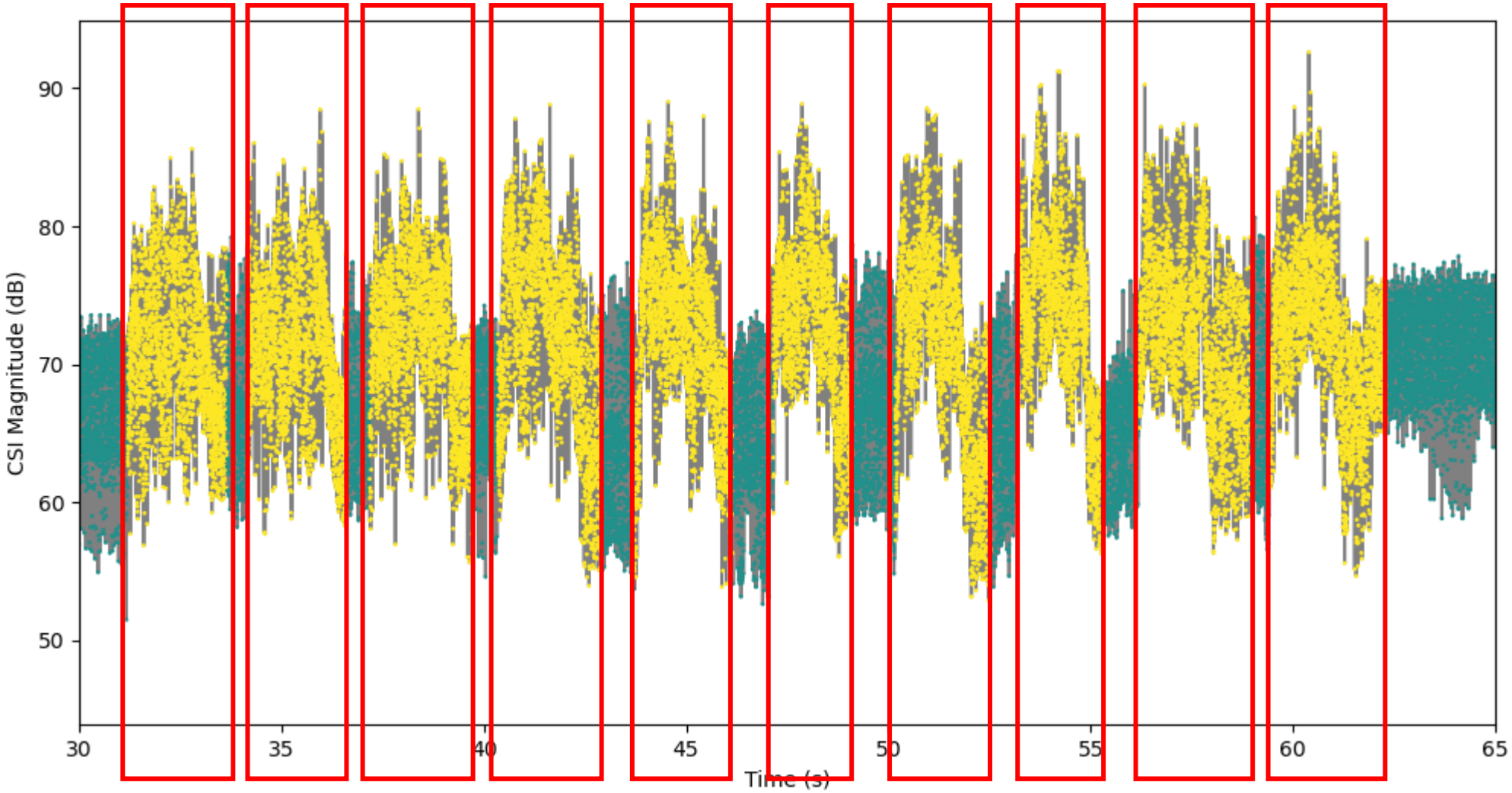
采集到的信号

难以理解

长短不一

动作识别

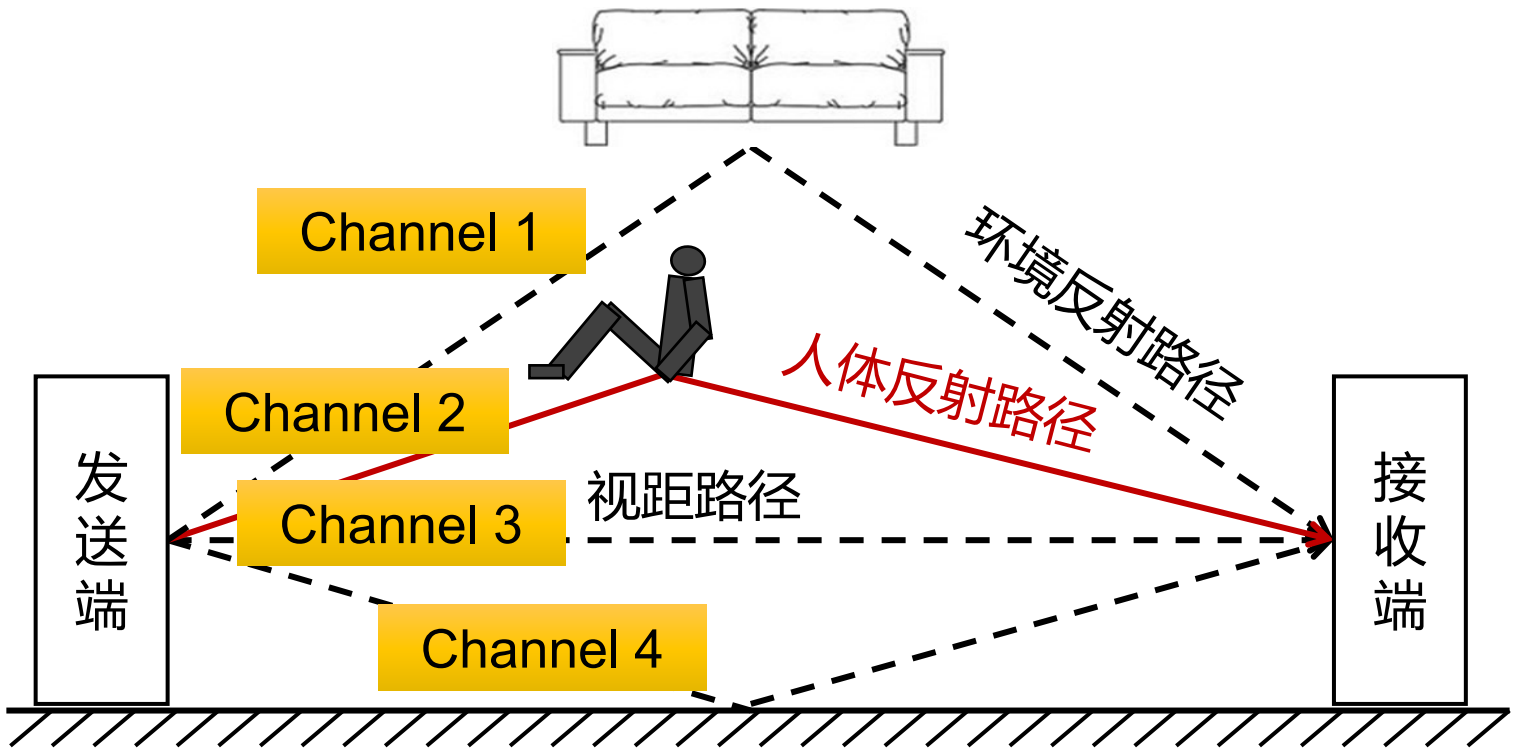




由于动作的不稳定性，采集到的信号长短不一

需要通过特征提取的方式进行结构化

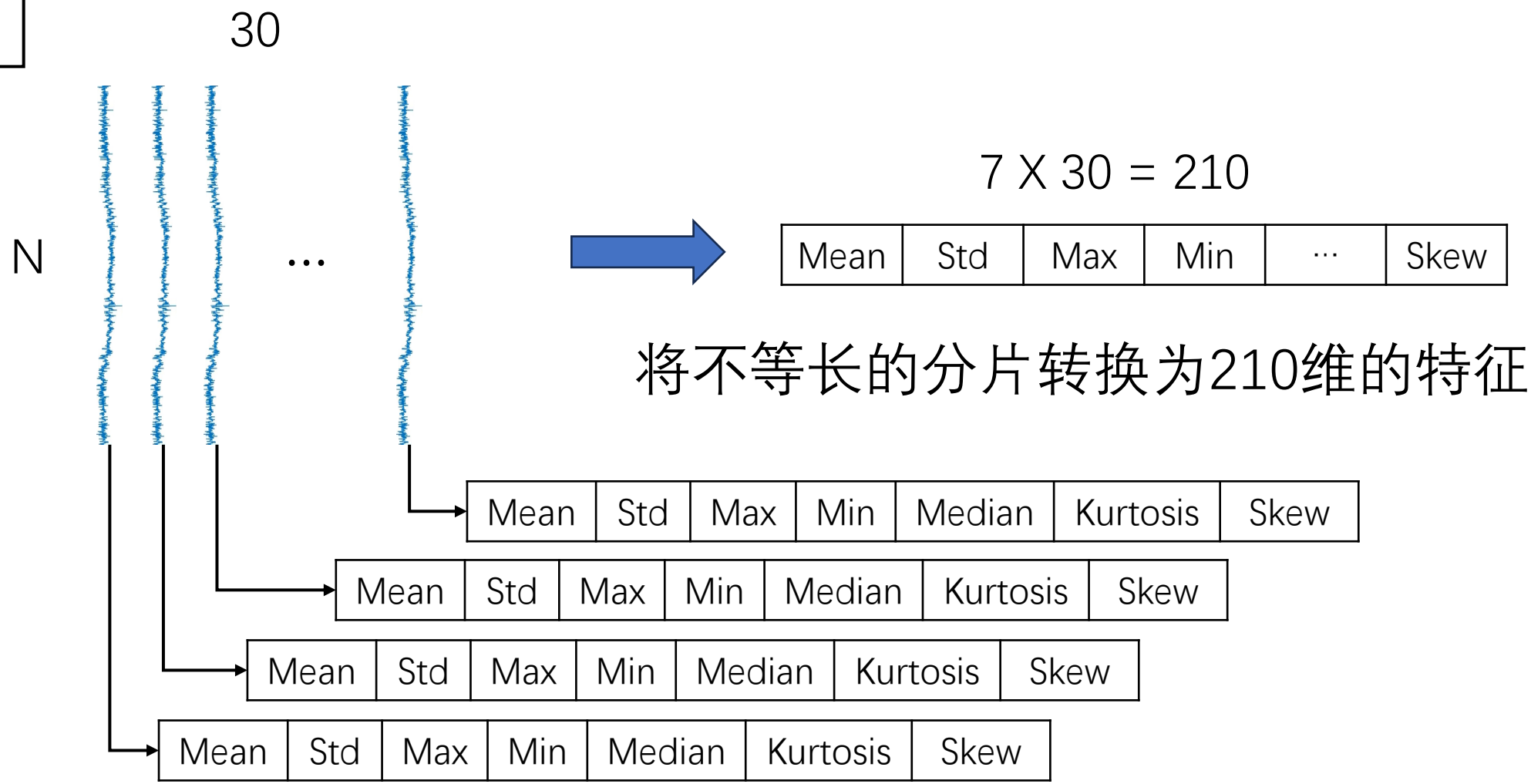
信道状态信息 Channel State Information（CSI）



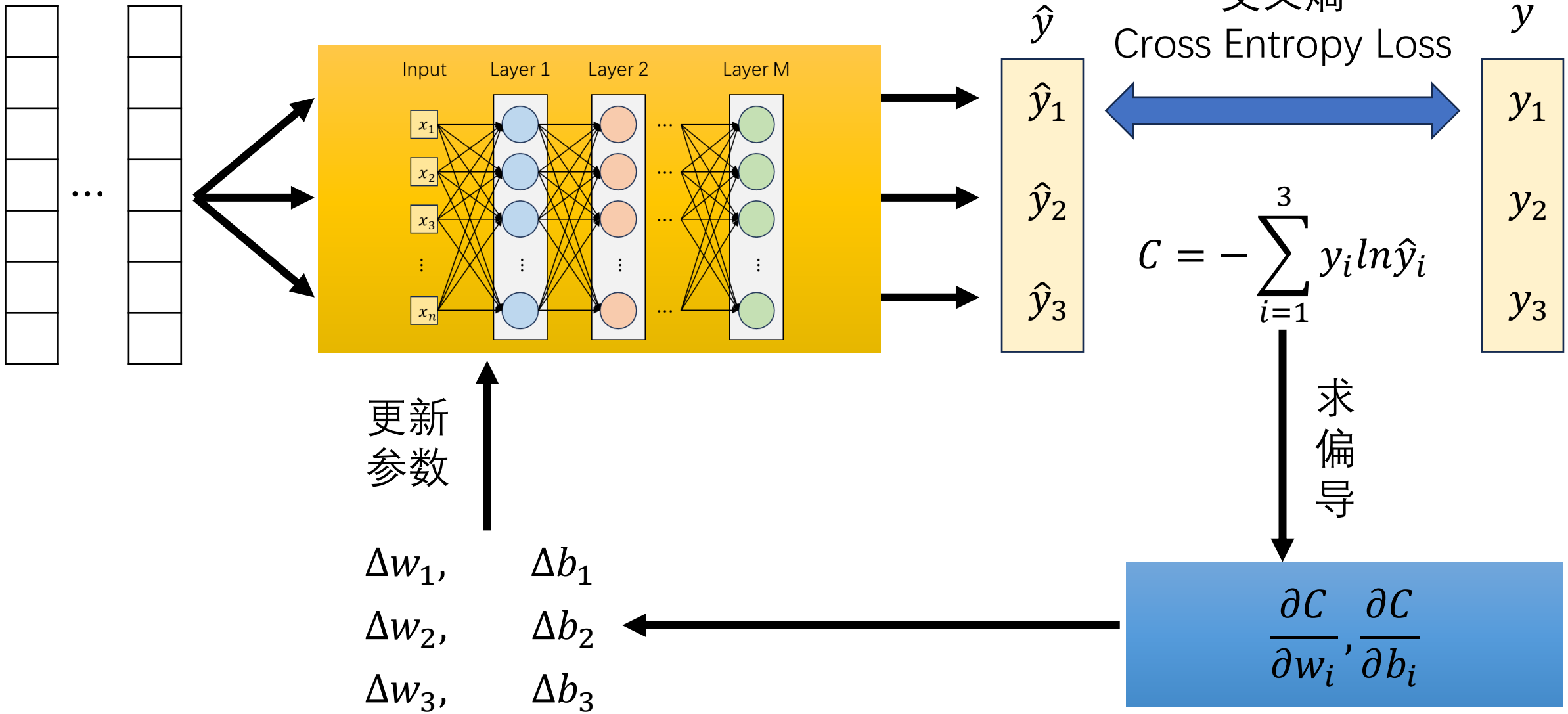
整体信号强度

CSI以较细的力度描述传播信道的情况

接收端



将不等长的分片转换为210维的特征向量


Input
Layer 1
Layer 2
Layer M
x_1
x_2
x_3
x_n
$\hat{y}$
$\hat{y}_1$
$\hat{y}_2$
$\hat{y}_3$
交叉熵
Cross Entropy Loss
y
y_1
y_2
y_3
$C = -\sum_{i=1}^{3} y_i ln\hat{y}_i$
求偏导
$\frac{\partial C}{\partial w_i}, \frac{\partial C}{\partial b_i}$
更新参数
$\Delta w_1, \quad \Delta b_1$
$\Delta w_2, \quad \Delta b_2$
$\Delta w_3, \quad \Delta b_3$